

75. If $x = 5 + 2\sqrt{6}$, then $\sqrt{x} - \frac{1}{\sqrt{x}}$ is (A.A.O. Exam, 2009)

- (a) $2\sqrt{2}$ (b) $2\sqrt{3}$
(c) $\sqrt{3} + \sqrt{2}$ (d) $\sqrt{3} - \sqrt{2}$

76. $(4 + \sqrt{7})$, expressed as a perfect square, is equal to (Section Officers', 2005)

- (a) $(2 + \sqrt{7})^2$ (b) $\left(\frac{\sqrt{7}}{2} + \frac{1}{2}\right)^2$
(c) $\left\{\frac{1}{2}(\sqrt{7} + 1)^2\right\}$ (d) $(\sqrt{3} + \sqrt{4})^2$

77. $\sqrt{8 - 2\sqrt{15}}$ is equal to (C.P.O., 2007)

- (a) $3 - \sqrt{5}$ (b) $\sqrt{5} - \sqrt{3}$
(c) $5 - \sqrt{3}$ (d) $\sqrt{5} + \sqrt{3}$

78. $\sqrt{6 - 4\sqrt{3}} + \sqrt{16 - 8\sqrt{3}}$ is equal to (A.A.O. Exam, 2010)

- (a) $1 - \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $2(2 - \sqrt{3})$ (d) $2(2 + \sqrt{3})$

79. The value of $\frac{1}{\sqrt{12 - \sqrt{140}}} - \frac{1}{\sqrt{8 - \sqrt{60}}} - \frac{2}{\sqrt{10 + \sqrt{84}}}$ is (S.S.C., 2005)

- (a) 0 (b) 1
(c) 2 (d) 3

80. The value of the expression

$$\sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}} - \sqrt{12 - 4\sqrt{5}}$$

- (a) an irrational number
(b) a negative integer
(c) a natural number
(d) a non-integer rational number

81. If $N = \frac{\sqrt{\sqrt{5} + 2} + \sqrt{\sqrt{5} - 2}}{\sqrt{\sqrt{5} + 1}} - \sqrt{3 - 2\sqrt{2}}$, then the value

of N is (A.A.O. Exam., 2009)

- (a) $2\sqrt{2} - 1$ (b) 3
(c) 1 (d) 2

82. Given that $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is close to

- (a) 1.45 (b) 1.88
(c) 2.9 (d) 3.7

83. If m and n are whole numbers such that $m^n = 121$, then the value of $(m - 1)^{n+1}$ is (S.S.C., 2001)

- (a) 1 (b) 10
(c) 121 (d) 1000

84. Number of prime factors in $(216)^{\frac{3}{5}} \times (2500)^{\frac{2}{5}} \times (300)^{\frac{1}{5}}$ is

- (a) 6 (b) 7
(c) 8 (d) None of these

85. Number of prime factors in $\frac{6^{12} \times (35)^{28} \times (15)^{16}}{(14)^{12} \times (21)^{11}}$ is

- (a) 56 (b) 66
(c) 112 (d) None of these

86. $1 + (3 + 1)(3^2 + 1)(3^4 + 1)(3^8 + 1)(3^{16} + 1)(3^{32} + 1)$ is equal to (Section Officers', 2005)

- (a) $\frac{3^{64} - 1}{2}$ (b) $\frac{3^{64} + 1}{2}$
(c) $3^{64} - 1$ (d) $3^{64} + 1$

87. $\frac{1}{1 + a^{(n-m)}} + \frac{1}{1 + a^{(m-n)}} = ?$ (M.B.A., 2003; NMAT, 2006)

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) a^{m+n}

88. If $a + b + c = 0$, then the value of $(x^a)^{a^2 - bc} \cdot (x^b)^{b^2 - ca} \cdot (x^c)^{c^2 - ab}$ is equal to

- (a) -2 (b) -1
(c) 0 (d) 1

89. $\frac{1}{1 + x^{(b-a)} + x^{(c-a)}} + \frac{1}{1 + x^{(a-b)} + x^{(c-b)}} + \frac{1}{1 + x^{(b-c)} + x^{(a-c)}} = ?$ (M.B.A., 2003)

- (a) 0 (b) 1
(c) $x^a - b - c$ (d) None of these

90. $\left(\frac{x^b}{x^c}\right)^{(b+c-a)} \cdot \left(\frac{x^c}{x^a}\right)^{(c+a-b)} \cdot \left(\frac{x^a}{x^b}\right)^{(a+b-c)} = ?$

(NMAT, 2005; L.I.C., 2003)

- (a) x^{abc} (b) 1
(c) $x^{ab+bc+ca}$ (d) x^{a+b+c}

91. $\left(\frac{x^a}{x^b}\right)^{(a+b)} \cdot \left(\frac{x^b}{x^c}\right)^{(b+c)} \cdot \left(\frac{x^c}{x^a}\right)^{(c+a)} = ?$ (M.B.A., 2006)

- (a) 0 (b) x^{abc}
(c) x^{a+b+c} (d) 1

92. $\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \cdot \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \cdot \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}} = ?$

- (a) 1 (b) $x^{\frac{1}{abc}}$
(c) $x^{\frac{1}{(ab+bc+ca)}}$ (d) None of these